

# **Medical Insurance Cost Prediction**

Using Linear Regression

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**Project Report**

# Executive Summary

This project develops a machine learning model to predict medical insurance charges using demographic and personal characteristics. The notebook implements Linear Regression with a Scikit-learn preprocessing pipeline. Categorical variables are one-hot encoded and numerical variables are passed through unchanged.

The model is evaluated using Mean Squared Error (MSE) and  $R^2$ . The recorded results are an MSE of **33,596,915.85** and an  $R^2$  score of **0.78**. The  $R^2$  result indicates that the model explains approximately 78% of the variation in test-set insurance charges.

## 1. Introduction

Medical insurance charges vary between individuals. Predictive modelling can be used to estimate these charges from available customer characteristics. This project applies a Linear Regression approach to demonstrate a complete machine learning workflow.

## 2. Problem Statement

The project aims to predict an individual's medical insurance charges from available demographic and personal attributes. The target variable is **charges**.

## 3. Objectives

The objectives are to prepare the insurance data, separate predictors from the target, encode categorical features, train a Linear Regression model, generate predictions for unseen test data, and evaluate the model using MSE and  $R^2$ .

## 4. Dataset

The notebook loads the Medical Insurance Dataset. The displayed data contains age, sex, bmi, children, smoker, region, and charges.

| Variable | Role / Description                |
|----------|-----------------------------------|
| age      | Numerical predictor               |
| sex      | Categorical predictor             |
| bmi      | Numerical predictor               |
| children | Numerical predictor               |
| smoker   | Categorical predictor             |
| region   | Categorical predictor             |
| charges  | Target: medical insurance charges |

## 5. Methodology

The notebook separates the features (X) from the target (y). The categorical variables are sex, smoker, and region, while age, children, and bmi are numerical. A ColumnTransformer applies OneHotEncoder(drop='first') to categorical variables and passes numerical variables through unchanged. The preprocessing and Linear Regression estimator are combined in a Scikit-learn Pipeline. The data is split into training and testing sets using test\_size=0.2 and random\_state=42.

## 6. Model

Linear Regression is used because the target, medical insurance charges, is continuous. The model provides an interpretable baseline for estimating charges from the selected predictors.

## 7. Model Evaluation

The recorded evaluation results are  $MSE = 33,596,915.85$  and  $R^2 = 0.78$ . An  $R^2$  of 0.78 indicates that approximately 78% of the variation in test-set charges is explained by the model. MSE represents the average squared prediction error.

## 8. Prediction Visualisation

The notebook creates an Actual vs Predicted Charges scatter plot. Actual charges are plotted against predicted charges, with a dashed reference line representing equality between actual and predicted values. This helps visually assess prediction accuracy.

## 9. Discussion

The project demonstrates a complete basic regression workflow: data loading, feature and target separation, categorical encoding, train-test splitting, pipeline construction, model fitting, prediction, and evaluation. The 0.78  $R^2$  score provides a useful baseline for this dataset.

## 10. Limitations

The project is limited to the variables and modelling approach implemented in the notebook. It does not document comparisons with alternative regression algorithms, cross-validation, or hyperparameter tuning. The Linear Regression result should therefore be treated as a baseline.

## 11. Future Improvements

Future work could compare Linear Regression with tree-based regression models, apply cross-validation, investigate feature engineering, analyse prediction errors, and compare multiple models using consistent metrics.

## 12. Conclusion

This project successfully implements a Linear Regression model for predicting medical insurance charges. On the recorded test evaluation, the model achieved an MSE of 33,596,915.85 and an  $R^2$  score of 0.78. The project provides a foundation for extending the analysis with more advanced modelling and validation techniques.